Clinical Study Summary Report (CSSR) Document Status: Draft | **Version:** 0.1 | **Date:** 03-Apr-2026

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1. Administrative & Study Identification

Full Study Title: A Phase 2 Open-label Randomized Study of V940 in Combination With BCG Versus BCG Monotherapy in Participants With High-risk Non-muscle Invasive Bladder Cancer (INTerpath-011) **Short Title/Acronym:** INTerpath-011 (V940-011) **Protocol Number:** V940-011 **NCT Number:** NCT06833073 **Trial Status:** Recruiting **Sponsor Name:** Merck Sharp & Dohme LLC **Investigational Product:** Intismeran autogene (V940/mRNA-4157) and Bacillus Calmette-Guérin (BCG vaccine) (ATC Code: L03AX03) **Phase of Development:** Phase II **Medical Monitor:** Merck Sharp & Dohme LLC Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the event-free survival (EFS) of intismeran autogene + BCG versus BCG alone in Cohort A, and to evaluate the complete response rate (CRR) by Blinded Independent Central Review (BICR) in Cohort B (Monotherapy Arm). **Secondary Objectives:** To evaluate 12-Month Event-free Survival (EFS), 24-Month Event-free Survival (EFS), recurrence-free survival (RFS), disease-specific survival, overall survival, time to cystectomy, and safety across cohorts. Additional secondary endpoints include Complete Response Rate (CRR) by BICR and duration of response (DOR) in participants with Carcinoma in situ (CIS).

2.2 Background & Rationale Transurethral resection of a bladder tumor (TURBT) followed by intravesical BCG is the standard of care for patients with treatment-naive high-risk (HR) non-muscle invasive bladder cancer (NMIBC). However, many patients develop disease recurrence and progression. Intismeran autogene (V940) is an individualized neoantigen therapy (mRNA vaccine) that is hypothesized to enhance the antitumor activity of BCG by eliciting endogenous antitumor T-cell responses directed at the unique neoantigens of the patient's tumor, providing a new targeted therapeutic pathway to improve long-term outcomes.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (BCG Monotherapy) **Blinding:** Open-label **Randomization:** 1:1 Randomization in Cohort A (BCG Arms); Cohort B (Monotherapy Arm) is a single-arm exploratory cohort.

3.2 Planned Enrollment & Duration Target N: Approximately 308 participants (278 in Cohort A; 30 in Cohort B) **Number of Sites:**

Multicenter / Global (approx. 89 locations) **Estimated Study Start:** 11-Mar-2025 **Estimated Primary Completion:** 03-Sep-2031

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Cohort A (BCG Arms): BCG-naive HR NMIBC (high-grade [HG] Ta, T1, and/or CIS). 3. Cohort B (Monotherapy Arm): CIS +/- papillary non-muscle invasive UC. BCG-naive or BCG-exposed HR NMIBC ineligible for or refusing intravesical therapy. HIV-infected individuals must have well-controlled HIV on antiretroviral therapy (ART). 4. Participants must have undergone a TURBT ≤ 12 weeks before randomization/allocation showing BICR-confirmed HR NMIBC histology.

4.2 Exclusion Criteria 1. Muscle-invasive bladder cancer, locally-advanced (T2, T3, T4), metastatic UC, or concurrent extravesical transitional cell carcinoma. 2. Current active tuberculosis or known history of HIV infection (for Cohort A), or HIV with Kaposi's sarcoma/Multicentric Castleman's Disease (for Cohort B). 3. Any contraindication to IV contrast and gadolinium, or inability to undergo imaging with CTU or MRU. 4. Has had a myocardial infarction within 6 months of randomization/allocation. 5. Has received prior treatment with a cancer vaccine or received any systemic anticancer therapy within 4 weeks. 6. Has immunodeficiency, is receiving chronic systemic steroid therapy, or has active autoimmune disease requiring systemic treatment in the last 2 years.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: - **Cohort A:** 50 mg intravesical BCG (induction: Q1W for 6 weeks; maintenance: Q1W for 3 weeks at weeks 13, 25, 49, and 73) $\pm$ Intismeran autogene 1 mg Q3W for 9 doses. - **Cohort B:** Intismeran autogene 1 mg Q3W for 9 doses. **Route of Administration:** BCG is administered intravesically; Intismeran autogene is administered intramuscularly (IM). **Duration of Treatment:** Intismeran autogene for approx. 27 weeks (9 doses); BCG for up to 73 weeks.

5.2 Concomitant & Prohibited Therapy Permitted: Standard concomitant medications for co-morbidities. **Prohibited:** Systemic anti-cancer treatments or other investigational agents within specified washout periods.

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6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :--- | :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent, Medical History, Tumor tissue and blood sample collection for NGS

and ctDNA testing | | **Baseline** | Day 0 | Randomization, First Dose of Intismeran/BCG | | **Interim** | Weeks 1-73 | Treatment Administration (Q1W for BCG induction/maintenance; Q3W for Intismeran) | | **Follow-up** | Every 12 / 24 weeks | Disease assessment (urine cytology by BICR and cystoscopy) every 12 weeks for the first 2 years, then every 24 weeks for years 3-5 |

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Event-free survival (EFS) for Cohort A (BCG Arms) and Complete Response Rate (CRR) by BICR for Cohort B (Monotherapy Arm). **Measurement Method:** Disease assessment utilizing urine cytology evaluated by Blinded Independent Central Review (BICR) and regular cystoscopy.

7.2 Secondary & Exploratory Endpoints - 12-Month Event-free Survival (EFS) and 24-Month Event-free Survival (EFS). - Recurrence-free Survival (RFS), disease-specific survival, and overall survival. - Time to cystectomy and overall safety (Adverse Events). - Complete Response Rate (CRR) by BICR and Duration of Response (DOR) for patients presenting with CIS at baseline.

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard continuous safety monitoring through clinical evaluations, laboratory assessments, and vital signs across all cohorts. **Serious Adverse Events (SAEs):** Captured and reported to the sponsor and applicable regulatory bodies within standard 24-hour reporting timelines. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee appointed to review safety and efficacy data periodically.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy assessments; Safety Set for participants receiving ≥ 1 dose of study treatment. **Sample Size Justification:** A sample size of roughly 308 patients (278 in Cohort A, 30 in Cohort B) is calculated to provide adequate power for detecting significant differences in EFS and complete response between the monotherapy and combination arms. **Handling of Missing Data:** Standard survival analysis methodologies (e.g., Kaplan-Meier estimation) with appropriate censoring rules for time-to-event endpoints.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits to ensure protocol adherence, drug accountability, and data integrity. **Audit Trail:** Validated Electronic Data Capture (EDC) system with full audit trails for eCRF data entry and query resolution.

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11. Study Identifiers & Governance

Primary ID: {{V940-011}} **Registry Number:** {{NCT06833073}} **Sponsor/Lead Organization:** {{Merck Sharp & Dohme LLC}} **Study Title:** {{A Clinical Study of Intismeran Autogene (V940) and BCG in People With Bladder Cancer (V940-011/INTerpath-011)}} **Official Regulatory Title:** {{A Phase 2 Open-label Randomized Study of V940 in Combination With BCG Versus BCG Monotherapy in Participants With High-risk Non-muscle Invasive Bladder Cancer (INTerpath-011)}}

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12. Clinical Framework (NMIBC Specific)

Primary Condition: {{Urinary Bladder Neoplasms, Non-Muscle Invasive Bladder Neoplasms, Carcinoma in situ (CIS)}} **Diagnosis Threshold:** {{Histologically confirmed HG Ta, T1, and/or CIS post-TURBT}} **Medical Methodology:** {{TURBT followed by Intravesical Therapy (BCG) and Individualized Neoantigen Therapy}} **Optimization Target:** {{Enhancement of endogenous antitumor T-cell responses directed at unique tumor neoantigens}}

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13. Study Procedures & Methodology

Management Pathway: {{Standard TURBT + BCG vs Combination with Intismeran Autogene (mRNA-4157)}} **Education Protocol (HCP):** {{Site initiation and protocol training for intravesical and intramuscular administration, and tissue handling for vaccine production}} **Education Protocol (Patient):** {{Informed consent outlining personalized therapy requirements and longitudinal follow-up}} **Quality Audit Mechanism:** {{Blinded Independent Central Review (BICR) for urine cytology and efficacy assessments}}

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14. Observation & Follow-Up Schedule

Total Duration: {{Up to 5 years of follow-up (until estimated completion in 2031)}} **Onsite Visit Cadence:** {{Every 12 weeks for the first 2 years, then every 24 weeks for years 3-5}} **Remote

Monitoring Frequency: {{Standard interval scheduling between dosing visits}} **Checklist Review Interval:** {{Prior to each Q3W Intismeran dosing cycle and corresponding BCG induction/ maintenance weeks}}

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]